

Benchmarking ^{136}Xe $0\nu\beta\beta$ NME calculations with the $^{138}\text{Ba}(p,t)$ reaction

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总问题

如何利用实验可观测量来检验 ^{136}Xe 无中微子双贝塔衰变核矩阵元 $M^{0\nu}$ 计算中与两核子配对转移有关的核结构输入？

① ^{136}Xe 的 $0\nu\beta\beta$ NME 是否存在模型依赖？

- 若 $0\nu\beta\beta$ 由 light Majorana neutrino exchange 驱动，衰变振幅与 $m_{\beta\beta}M^{0\nu}$ 成正比。
- $M^{0\nu}$ 可分解为 Gamow-Teller、Fermi 和 Tensor 部分：

$$M^{0\nu} = M_{GT}^{0\nu} - \left(\frac{g_V}{g_A}\right)^2 M_F^{0\nu} + M_T^{0\nu}.$$

- 其中 $M_{GT}^{0\nu}$ 通常是主导项。
- 对 ^{136}Xe ，不同模型，如 QRPA、ISM、CDFT、EDF、IBM-2、GCM，给出的 $M^{0\nu}$ 差异明显，说明需要实验核结构信息进行 benchmark。

- ① 壳模型中的 $J^\pi = 0^+$ **pair-transfer** 贡献是否被低估？
- 在壳模型或 QRPA 中, $M^{0\nu}$ 可按中间核态或核子对角动量分解。
 - 对 ^{136}Xe , $J^\pi = 0^+$ neutron-pair removal 相关部分对 Gamow-Teller NME 有重要正贡献。
 - $J > 0$ 项通常会部分抵消 $J = 0$ 项, 因此 $J = 0$ 部分的大小直接影响完整 NME 的模型判断。
 - 有限价空间壳模型可能遗漏价空间外轨道的相干贡献, 从而低估 $J^\pi = 0^+$ pair-transfer amplitude。

- ② (p, t) 两中子转移反应能否检验相关 TNA？

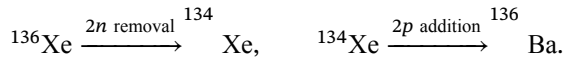
- (p, t) 反应对应两中子移除:

$$^A Z(p, t)^{A-2} Z.$$

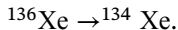
- 该反应对 $L = 0$ neutron-pair transfer 特别敏感, 因而可以探测 even-even 核中的 pair-correlated 0^+ 态。
- 实验角分布和截面可以与 shell-model TNA + FRESCO 反应计算比较, 从而检验理论波函数中的 two-neutron transfer amplitudes。

① 为什么选择 $^{138}\text{Ba}(p, t)^{136}\text{Ba}$?

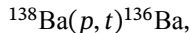
- ^{136}Xe 的 NME 可形式上写成经由 $A-2$ 中间核态的求和:



- 直接的两中子移除端是



- 文章选择类比反应

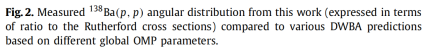


因为 ^{136}Xe 与 ^{138}Ba 都位于 $N=82$ 附近, 具有近球形、单闭壳特征。

- 同时, ^{134}Xe 与 ^{136}Ba 的低能级结构相似, 因此 $^{138}\text{Ba}(p, t)$ 可作为 $^{136}\text{Xe} \rightarrow ^{134}\text{Xe}$ 两中子移除过程的 benchmark。

本文的一句话目标

利用 $^{138}\text{Ba}(p, t)^{136}\text{Ba}$ 实验数据检验壳模型 two-neutron transfer amplitudes, 并进一步约束 $^{136}\text{Xe} 0\nu\beta\beta$ NME 中 $J^\pi = 0^+$ Gamow-Teller 贡献。



- $^{138}\text{Ba}(\text{p,p})$ 弹散，把实验弹散角分布和不同全局 OMP 的 DWBA 计算比 $\Rightarrow$ Varner OMP R. Varner et al. Phys. Rep. 201, 57 (1991)

- $^{136}\text{Ba}+t \Rightarrow \text{Li OMP}$ X. Li et al. Nucl. Phys. A 789, 103 (2007)
- core-2n coupling OMP: 其井深经过调整, 以重现每个中子的结合能。 F. D. Becchetti and G. W. Greenlees, Phys. Rev. 182, 1190 (1969)
- $^{137}\text{Ba}(d,d) \Rightarrow \text{An,Cai OMP}$ H. An and C. Cai, Phys. Rev. C 73, 054605 (2006)

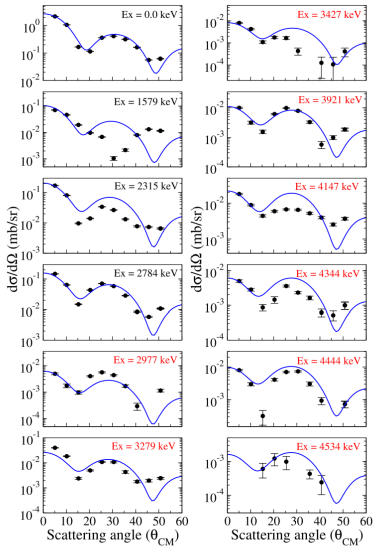


Fig. 3. Angular distributions of 0^+ states identified in this work. The solid curves are normalized DWUCK4 DWBA predictions for $L=0$ transfer, assuming a $(0h_{11/2})^2$

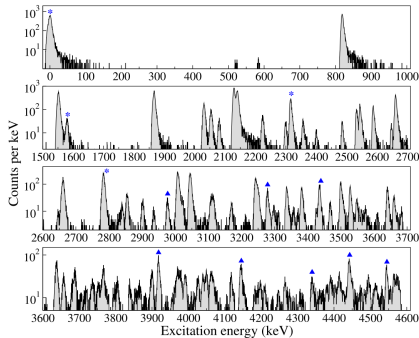


Fig. 1. Excitation energy spectrum in ^{136}Ba obtained at $\theta_{\text{lab}} = 25^\circ$. Previously known 0^+ states are marked with asterisks, while the new 0^+ states observed in this work are shown with filled triangles.

- 每个小图对应 ^{136}Ba 中一个 0^+ 态
- 由角分布形状 $\rightarrow L=0$ 判定 $\rightarrow 0^+$

Table 2

Measured cross sections at $\theta_{lab} = 5^\circ$ and relative (p, t) strengths obtained from the data shown in Fig. 3.

E_x [keV]	$(d\sigma/d\Omega)_{5^\circ}$ [mb/sr]	ϵ_i [%]
0	2.17(12)	100.0
1579	0.071(4)	5.1(7)
2315	0.17(1)	15.2(19)
2784	0.148(8)	14.6(17)
2977	0.0046(6)	0.65(9)
3279	0.041(2)	3.3(3)
3427	0.0082(8)	1.1(1)
3921	0.0096(8)	2.2(3)
4147	0.018(1)	5.4(7)
4344	0.0055(6)	1.8(3)
4444	0.0075(7)	3.2(4)
4534 ^a	...	0.6(3)
Integrated $L = 0$ strength relative to the ground state		$\sum \epsilon_i = 53(3)\%$

- 相对强度

$$\epsilon_i = \left[\frac{\left(\frac{d\sigma}{d\Omega} \right)_{0^+ex}^{data}}{\left(\frac{d\sigma}{d\Omega} \right)_{0^+ex}^{DWBA}} \right]_i \left[\frac{\left(\frac{d\sigma}{d\Omega} \right)_{G.S.}^{DWBA}}{\left(\frac{d\sigma}{d\Omega} \right)_{G.S.}^{data}} \right], \quad (1)$$

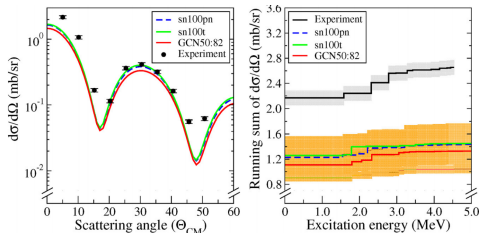
对每个激发 0^+ 态，用 DWBA 归一化到前角数据，得到一个归一化因子；再除以基态的归一化因子，得到相对基态的 pair-transfer strength.

- why? 因为不同激发态有不同 Q 值、运动学因素和 DWBA 几何因素。直接拿实验截面相除不够干净。用 DWBA 归一化因子可以尽量去掉这些运动学影响。

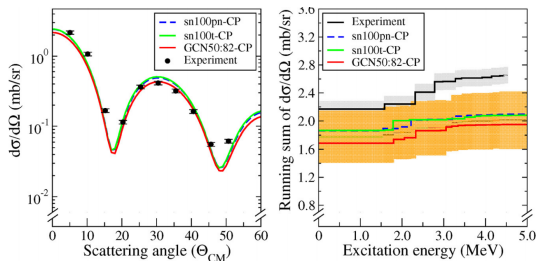
The results from this experiment also allow tests of the nuclear structure models that are used to calculate the NME for ^{136}Xe $0\nu\beta\beta$ decay. We performed one such test using configuration interaction shell model calculations with the NuShellX code [63]. Unlike the DWUCK4 calculations which merely served to identify the observed 0^+ states and determine relative strengths, the shell model calculations were used to assess the absolute (p, t) cross sections, distributed over 0^+ states in ^{136}Ba . The calculations used the five-orbital $(0g_{7/2}, 1d_{5/2}, 1d_{3/2}, 2s_{1/2}, 0h_{11/2})$ valence space for protons and neutrons to determine the wavefunctions for the 0^+ ground state of ^{138}Ba and the lowest fifty 0^+ states in ^{136}Ba . We next used NuShellX to calculate two-neutron transfer amplitudes between these states. The amplitudes were input to the Fresco [64] coupled-reaction channels code, which was used to generate absolute $^{138}\text{Ba}(p, t)$ angular distributions. The Fresco calculations used the same OMP parameters for the proton and triton channels (as described previously) and also took into account the coherent sum of both direct and sequential two-step transfer. The single-nucleon transfer amplitudes for the two-step part were obtained assuming one intermediate state in ^{137}Ba for each of the transferred (n, ℓ, j) values. For the ^{137}Ba -deuteron coupling, we used the global OMP parameters recommended by An and Cai [65], based on a comparison with experimental $^{138}\text{Ba}(d, d)$ angular distribution data that we obtained independently [66].

- ① 五轨道价空间: $0g_{7/2}, 1d_{5/2}, 1d_{3/2}, 2s_{1/2}, 0h_{11/2}$
- ② $\text{NuShellX} \Rightarrow \Psi(^{138}\text{Ba}; 0_{g.s}^+), \Psi(^{136}\text{Ba}; 0_i^+) \Rightarrow \text{TNA}$
- ③ 用 fresco 计算:
 - 顺序转
移: $p + ^{138}\text{Ba} \rightarrow d + ^{137}\text{Ba} \rightarrow t + ^{136}\text{Ba}$
 - 同时转移: $p + ^{138}\text{Ba} \rightarrow t + ^{136}\text{Ba}$
 - 相干部分

We used three different Hamiltonians for the calculations, which were corrected for core-polarization due to configuration mixing with orbitals outside the model space [67]. The first Hamiltonian is from Ref. [68] and is called sn100pn in the NuShellX interaction library [63]. The second Hamiltonian (that we call sn100t) is very similar to sn100pn, except with minor modifications and was used in Ref. [12] to calculate $M^{0\nu}$ for ^{136}Xe $\beta\beta$ decay, while the third GCN50:82 [69] Hamiltonian was used by the authors of Ref. [35] to calculate the NME for the decay. Our results are discussed below.



- 壳模型计算出的 TNA 和 (p,t) 截面结果，不是强烈依赖某一套相互作用
- 有限价空间能够定性预测基态转移占主导，但在前角绝对截面上低估实验，最大约低 2 倍。



- ladder-diagram 修正 core-polarization: 考虑中子对散射到模型空间外的 23 个轨道, 最高到 $i_{11/2}$ 。P. Decowski et al. Nucl. Phys. A 302, 186 (1978)
- 加入这个修正后, 预测的 (p,t) 截面增强约 1.5 倍, 和实验更接近。

$$R = \frac{M_{GT}^{0\nu}(J^\pi = 0^+) \text{ with core-polarization}}{M_{GT}^{0\nu}(J^\pi = 0^+) \text{ without core-polarization}}$$

In the final part of our analysis we used these results to benchmark NME calculations for ^{136}Xe $0\nu\beta\beta$ decay. This was based on the arguments presented in Ref. [71], where it was shown that the $0\nu\beta\beta$ decay NME for a parent nucleus with mass number A can be expanded as a sum over states in an intermediate nucleus with mass number $(A-2)$. For the case of ^{136}Xe , one can similarly evaluate the NME by summing over the products of the TNA for two-neutron removal to ^{134}Xe , the TNA for two-proton addition to ^{136}Ba , and the two-body matrix element for the double-beta decay operators (cf. Eq. (9) in Ref. [71]). The most significant contribu-

We will analyze the $0\nu\beta\beta$ matrix elements in terms of their dependence on the intermediate states

$$M^{0\nu}(E_x, J_m) = \sum_{E_m < E_x, J_m} V(f, i, m),$$

where

$$V(f, i, m) = \sum_{k_\alpha \leq k_\beta, k_\gamma \leq k_\delta} \langle k_\alpha, k_\beta, J_m | V | k_\gamma, k_\delta, J_m \rangle \\ \times \text{TNA}(f, m, k_\alpha, k_\beta, J_0) \text{TNA}(i, m, k_\gamma, k_\delta, J_0).$$

B. A. Brown et al. PRL 113, 262501 (2014)

why choose $^{138}\text{Ba}(p,t)^{136}\text{Ba}$

- ^{136}Xe 和 ^{138}Ba 都是单闭壳，近球形核， $N=82$
- ^{134}Xe 和 ^{136}Ba 的低能级结构几乎相同，理论计算预测它们有相似的基态结构。

nuclei, their 2_1^+ states have very similar excitation energies and $B(E2; \uparrow)$ values [73]. Additional comparison, after including recent results from $(n, n'\gamma)$ experimental work [74], shows that the energies of the 2_2^+ , 2_3^+ , 0_2^+ , 4_1^+ , 4_2^+ and 7_1^- states are also very similar in both nuclei. Therefore, the low-lying level schemes in ^{134}Xe and ^{136}Ba are nearly identical. This similarity allows a benchmarking of ^{136}Xe $0\nu\beta\beta$ decay NME calculations using our $^{138}\text{Ba}(p,t)$

① $J^\pi = 0^+$ Gamow-Teller NME 分量的修正：
计算了 ^{134}Xe 种 $J^\pi = 0$ 基态的 NME, 分别使用加入和不加入 core-polarization 修正的 TNA。得到修正因子 $R = 1.58$

② 之后作者进行：五轨道价空间 ISM NME 计算，使用 sn100t Hamiltonian 采用 CD-Bonn 势处理短程关联，并加入一些高阶贡献后，得到矩阵元
 $M_{GT}^{0\nu}(J^\pi = 0^+) = 5.67$

$$M_{GT}^{0\nu}(J^\pi = 0^+) = R \times 5.67 = 8.96.$$

接近 QRPA 计算的 $J = 0\text{GT}$ 矩阵元：

$$M_{GT}^{0\nu}(J^\pi = 0^+)_{\text{QRPA}} = 9.63$$

- Shell-model TNA were calculated with NuShellX.
- These TNA were used in FRESKO to calculate absolute (p, t) cross sections.
- Without core-polarization corrections, the calculated cross sections underestimate the data by about a factor of two.
- Including pair-scattering contributions from outside the valence space enhances the calculated cross sections and improves agreement with experiment.
- The correction enhances the shell-model estimate of

$$M_{GT}^{0\nu}(J^\pi = 0^+)$$

from 5.67 to 8.96.

Important limitation

This benchmark constrains only the $J^\pi = 0^+$ GT component, not the full $0\nu\beta\beta$ NME.

Thanks!